DP Biology – D4.1 (HL extension)

# Selection and gene pools

- What can change allele frequencies in a population?
- What part does reproduction play in natural selection?

HL statements D4.1.9–15. Textbook pp. 790–797.
Image: a male guppy. 5snake5, CC0.



# Objectives for each class

| Class | Statements | By the end of the class, you can... |
|---|---|---|
| **1** | D4.1.9–10 | Describe gene pools, and compare allele frequencies in separated populations |
| **2** | D4.1.11–12 | Explain how selection changes allele frequencies, and compare three patterns of selection |
| **3** | D4.1.13–14 | Use the Hardy-Weinberg equation and explain its conditions |
| **4** | D4.1.15 | Explain artificial selection and how it differs from natural selection |

# The gene pool



- A **gene pool** is all the genes and all their alleles in an interbreeding population
- **Allele frequency** is the proportion of all copies of a gene that are one allele
- It is written as a decimal between 0 and 1
- Evolution is a change in allele frequencies over generations

### Worked example

100 flowers: 49 RR, 42 Rr, 9 rr.
Copies of the gene: 200.
Copies of R: 98 + 42 = 140.
Frequency of R = 140 / 200 = 0.7.
Frequency of r = 0.3.

### Key idea

Each population has its own gene pool, because mating happens mainly within it.

# Allele frequencies differ between populations

D4.1.10

| Allele (approximate frequency) | Northern Europe | East Asia | West Africa |
|---|---|---|---|
| **Lactase persistence, -13910T** | 0.75 | Close to 0 | Close to 0 |
| **ALDH2 variant rs671 (alcohol flush)** | Close to 0 | 0.25 | Close to 0 |
| **Sickle cell, HbS** | Close to 0 | Close to 0 | Up to 0.15 |

Rounded values from published population surveys. Frequencies vary within each region.

- Separated populations have different mutations, selection pressures and chance events
- Example: malaria selects for HbS carriers where the disease is common

**Research it**

Look up one allele in a public database such as dbSNP or Ensembl, and compare its frequency in two populations.

# Selection changes allele frequencies

D4.1.11



Peppered moth, pale form. Chiswick Chap, CC BY-SA 3.0.

- Individuals vary in heritable traits
- Some survive and reproduce more than others
- They pass on more copies of their alleles, so those alleles rise in frequency
- Combining Darwin with genetics is called **neo-Darwinism**

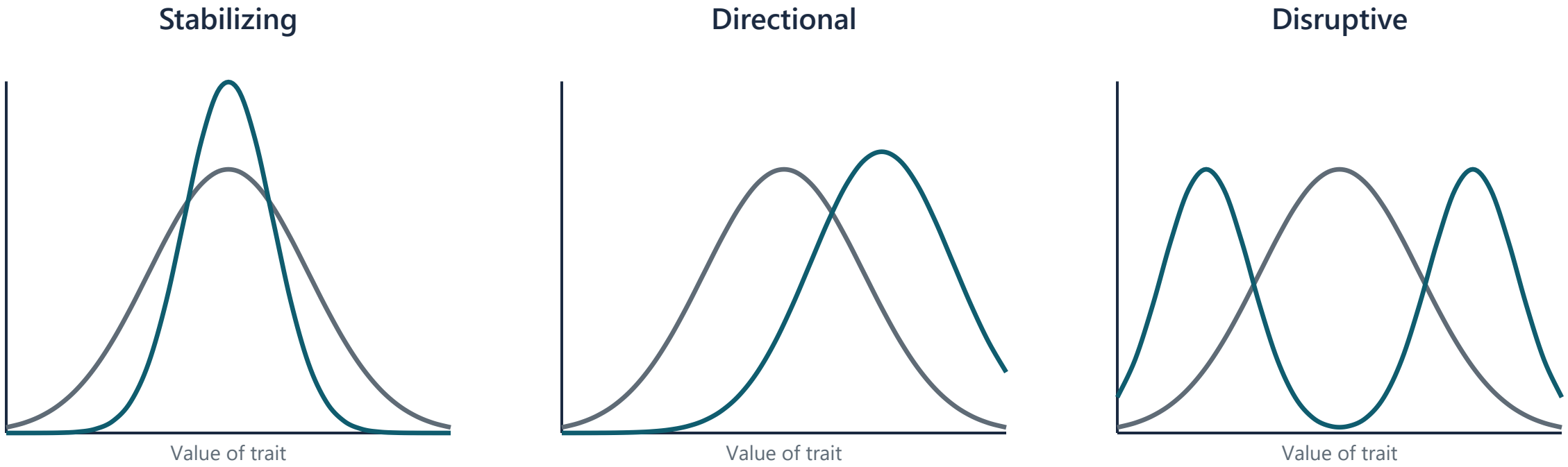
Soot darkens tree trunks → Dark moths are better hidden from birds → More dark moths survive to breed → The dark allele rises in frequency

### Reversal

After clean air laws, pale moths were better hidden again and the dark allele fell in frequency.

# Three patterns of selection

D4.1.12




Grey: population before selection. Teal: population after selection.

| Favours the mean | Favours one extreme | Favours both extremes |
|---|---|---|
| **Human birth weight** | Dark moths; larger turkeys | Finches with very large or small beaks |

All three can change allele frequencies; stabilizing selection removes alleles that give extreme phenotypes.

# The Hardy-Weinberg equation

D4.1.13

$$p + q = 1$$

$$p^2 + 2pq + q^2 = 1$$

| Symbol | Meaning | Who has it |
|---|---|---|
| **p, q** | Frequencies of two alleles | p dominant, q recessive |
| **$p^2$** | Genotype frequency | Homozygous dominant |
| **2pq** | Genotype frequency | Heterozygous |
| **$q^2$** | Genotype frequency | Homozygous recessive |

**Tip**

Start with the homozygous recessive group, as it is the only genotype you can see directly from the phenotype.

# Worked example: albinism



1. About 1 person in 20 000 has albinism (aa)
2. $q^2 = 1 / 20\,000 = 0.00005$
3. q = square root of 0.00005 = 0.007
4. p = 1 − q = 0.993
5. Carriers: $2pq = 2 \times 0.993 \times 0.007 = 0.014$

### Meaning

About 1 person in 70 is a carrier, even though only 1 in 20 000 has the condition. Most copies of a rare recessive allele are hidden in carriers.

### Try it

Tay-Sachs disease (tt) has a frequency of 0.0003 in one population. Calculate q and the frequency of carriers.

# Conditions for Hardy-Weinberg equilibrium

D4.1.14

| Condition | What breaks it |
| --- | --- |
| **Random mating** | Choosing mates by phenotype |
| **No selection** | Some genotypes survive or reproduce better |
| **No mutation** | New alleles arise |
| **No migration** | Individuals move in or out with different alleles |
| **Very large population** | Chance changes in small populations (genetic drift) |

### Using it

If observed genotype frequencies do not fit the equation, at least one condition is not being met, so something is changing the population.

# Artificial selection

D4.1.15



Two dogs, one species: selective breeding. Elf, CC BY-SA 3.0.



Bread wheat. Bluemoose, CC BY-SA 3.0.

- Dogs from wolves
- Wheat with larger grains

| | Natural selection | Artificial selection |
|---|---|---|
| **What selects** | The environment | People, deliberately |
| **Favours** | Traits that aid survival and reproduction | Traits people want, even if harmful to the organism |

**Careful**

Antibiotic resistance is natural selection: nobody chose it.

# Check-in: gene pools

D4.1.11–16

1. Define the gene pool of a population.
2. Explain how natural selection changes allele frequencies.
3. Distinguish between stabilizing, directional and disruptive selection.
4. State the Hardy-Weinberg equations and what p and q stand for.
5. List the conditions for Hardy-Weinberg equilibrium.
6. Distinguish between natural and artificial selection.

**How to use this**

Answer without your notes, then compare with a partner.

# Working with data: Hardy-Weinberg practice

D4.1.14

In a population of 1000 snails, 160 have yellow shells, the recessive phenotype (aa). The rest are brown. These numbers are invented for practice.

1. Calculate q, the frequency of the recessive allele.
2. Calculate p.
3. Calculate the expected number of heterozygous snails.
4. Bird predators eat more yellow snails. Predict how q will change and name the type of selection.

# Common mistakes to avoid

| Not this | But this |
| --- | --- |
| **q = number of recessive individuals / total.** | $q^2$ = that fraction; q is its square root. |
| **Dominant alleles always become more common.** | Allele frequencies change only through selection, drift, mutation or migration. |
| **Hardy-Weinberg describes real populations exactly.** | It is a model; differences from it show evolution. |

# Check that you can...

1. **Define** gene pool and allele frequency
2. **Explain** how selection changes allele frequencies
3. **Distinguish** between stabilizing, directional and disruptive selection
4. **Calculate** allele and genotype frequencies with the Hardy-Weinberg equations
5. **Outline** the conditions for Hardy-Weinberg equilibrium
6. **Compare** natural and artificial selection

# Exit activity: 5 minutes

**On your own and without notes. Hand this in as you leave.**

1. Define gene pool.

2. In a population, 4% are homozygous recessive. Calculate the frequency of the dominant allele.

3. State why antibiotic resistance is not an example of artificial selection.

# Image credits

Poecilia reticulata – Guppy, male, cobra-green morph (cropped).jpg – 5snake5 – CC0

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Big and little dog.jpg – Elf at English Wikipedia – CC BY-SA 3.0

Wheat close-up.JPG – User:Bluemoose – CC BY-SA 3.0

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